

ARKA-ED: Production-Ready Medical Education Platform

Complete Cursor Build Guide — From Zero to Launch

Version: 2.0

Build Tool: Cursor IDE

Tech Stack: Next.js 14, TypeScript, Tailwind CSS, Supabase, Framer Motion

Estimated Build Time: 16-20 hours

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1. Executive Vision

The \$2.15B Market Void

Every existing radiology education platform teaches the wrong skill.

Platform	Focus	ARKA-ED Advantage
Radiopaedia	Image interpretation	We teach WHEN to order
STATdx	Radiologist support	Built for ordering physicians
RADPrimer	Board exam prep	Practical ordering skills
UWorld	USMLE questions	Deep appropriateness curriculum
ACR TEACHES	Free 30-min modules	Interactive simulation

Critical Statistics:

- Only **2.4%** of physicians use ACR Appropriateness Criteria
- Over **50%** of residents unfamiliar with guidelines
- **\$100B+** wasted annually on inappropriate imaging

What We're Building

A case-based simulation platform where learners:

1. Receive clinical vignettes
2. Make imaging ordering decisions
3. Get instant ACR feedback (1-9 ratings)
4. Learn WHY with evidence-based explanations
5. See cost & radiation consequences
6. Track progress across competencies

Demo Scope

- 50+ clinical cases across 5 categories
- 5 specialty tracks (EM, IM, FM, Surgery, Peds)
- Full assessment engine with quizzes
- Gamification (streaks, achievements)

- Professional admin panel
 - Mobile-responsive design
-

Phase 1: Foundation

Prompt 1: Project Initialization

I'm building ARKA-ED, a medical education platform teaching imaging appropriateness. Create a Next.js 14 project with the complete structure below.

Run these commands first:

```
```bash
```

```
npx create-next-app@latest arka-ed-demo --typescript --tailwind --eslint --app --src-dir --import-alias "@/*"
cd arka-ed-demo
```

```
npm install framer-motion lucide-react @supabase/supabase-js @supabase/ssr zustand react-hook-form
@hookform/resolvers zod date-fns recharts clsx tailwind-merge class-variance-authority @radix-ui/react-dialog
@radix-ui/react-dropdown-menu @radix-ui/react-tabs @radix-ui/react-tooltip @radix-ui/react-checkbox @radix-
ui/react-progress @radix-ui/react-select @radix-ui/react-switch canvas-confetti
```

```
npm install -D @types/canvas-confetti @tailwindcss/typography @tailwindcss/forms
```
```

Create this complete folder structure with placeholder files:

```
src/
├── app/
│   ├── (auth)/
│   │   ├── login/page.tsx
│   │   ├── register/page.tsx
│   │   ├── forgot-password/page.tsx
│   │   ├── onboarding/page.tsx
│   │   └── layout.tsx
│   ├── (dashboard)/
│   │   ├── cases/
│   │   └── page.tsx
```

```

| | | | [caseId]/page.tsx
| | | | progress/page.tsx
| | | | assessments/
| | | | | page.tsx
| | | | | create/page.tsx
| | | | | [assessmentId]/page.tsx
| | | | specialty/[track]/page.tsx
| | | | achievements/page.tsx
| | | | layout.tsx
| | | (marketing)/
| | | | page.tsx
| | | | pricing/page.tsx
| | | | layout.tsx
| | | admin/
| | | | page.tsx
| | | | cases/page.tsx
| | | | users/page.tsx
| | | | layout.tsx
| | | api/auth/callback/route.ts
| | | layout.tsx
| | | globals.css
| | | not-found.tsx
| | | error.tsx
| | components/
| | | ui/ (button, card, input, badge, modal, tabs, progress, select, checkbox, switch, tooltip, skeleton, toast,
dropdown-menu, empty-state, error-state, loading-state)
| | | cases/ (case-card, case-grid, case-filters, case-viewer, clinical-vignette, patient-card, ordering-
interface, imaging-option-card, feedback-panel, acr-rating-badge, acr-rating-scale, radiation-indicator, cost-
display, learning-mode-toggle, hint-system, clinical-pearl)
| | | assessments/ (assessment-card, quiz-timer, quiz-progress, quiz-builder, results-breakdown)
| | | progress/ (stats-card, progress-chart, competency-radar, category-bar, streak-calendar, activity-feed,
achievement-badge)
| | | specialty/ (track-header, curriculum-module, track-progress)
| | | layout/ (navbar, footer, dashboard-sidebar, dashboard-header, dashboard-shell, admin-sidebar, mobile-
nav)
| | | marketing/ (hero, feature-card, stats-bar, process-flow, testimonial-card, pricing-card, cta-section,
comparison-table)
| | | auth/ (auth-form, social-buttons, auth-guard, role-guard)

```

```

|   |—— admin/ (case-editor, case-table, user-table)
|   |—— animations/ (fade-in, slide-in, stagger-children, page-transition, confetti)
|—— lib/
|   |—— supabase/ (client.ts, server.ts, middleware.ts, admin.ts)
|   |—— hooks/ (use-user.ts, use-auth.ts, use-cases.ts, use-progress.ts, use-assessment.ts, use-toast.ts, use-
learning-mode.ts)
|   |—— utils.ts
|   |—— constants.ts
|   |—— validations.ts
|   |—— calculations.ts
|—— types/ (database.ts, cases.ts, assessments.ts, progress.ts, index.ts)
|—— data/ (cases/, imaging-options.ts, specialty-tracks.ts, achievements.ts)
|—— stores/ (case-store.ts, user-store.ts, assessment-store.ts, ui-store.ts)
|—— providers/ (toast-provider.tsx, auth-provider.tsx)

```

Each file should have proper TypeScript interfaces and a placeholder component/function. Create a comprehensive README.md with setup instructions.

Prompt 2: Supabase Configuration

Set up Supabase configuration for ARKA-ED with browser, server, and middleware clients.

1. Create .env.local.example:

```

NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_anon_key
SUPABASE_SERVICE_ROLE_KEY=your_service_role_key
NEXT_PUBLIC_APP_URL=http://localhost:3000

```

2. src/lib/supabase/client.ts

Browser-side client using createBrowserClient from @supabase/ssr with Database generic typing.

3. src/lib/supabase/server.ts

Server-side client for Server Components using createServerClient with Next.js cookies() handling. Include getUser() and getProfile() helpers.

4. src/lib/supabase/middleware.ts

Export updateSession function for auth session refresh with proper cookie handling.

5. src/middleware.ts (project root)

Protect routes:

- Protected: /cases/*, /progress/*, /assessments/*, /specialty/*, /achievements/*
- Admin: /admin/* (check role)
- Public: /, /login, /register, /pricing
- Redirect unauthenticated to /login?redirect=path
- Redirect authenticated away from auth pages to /cases

6. src/providers/auth-provider.tsx

Auth context with user state, loading state, signIn, signUp, signOut, and onAuthStateChange listener.

Include proper TypeScript types, error handling, and JSDoc comments throughout.

Prompt 3: Design System & Tailwind

Create a distinctive medical education design system for ARKA-ED. Avoid generic AI aesthetics.

1. Update tailwind.config.ts with:

COLORS:

- primary: Deep navy (slate-900 base, #0f172a main)
- accent: ARKA Cyan (#06b6d4, sky-500 family)
- appropriate (ACR 7-9): Emerald (#10b981)
- maybe (ACR 4-6): Amber (#f59e0b)
- inappropriate (ACR 1-3): Rose (#f43f5e)
- specialty colors: em (red), im (blue), fm (green), surgery (purple), peds (teal)

FONTS:

- sans: Geist Sans or Inter
- display: Plus Jakarta Sans or Cal Sans
- mono: Geist Mono

ANIMATIONS (keyframes):

- fade-in, fade-up, fade-down

- slide-in-right, slide-in-left
- scale-in
- shimmer (for skeletons)
- pulse-glow
- bounce-subtle
- count-up
- draw-check

SHADOWS:

- soft: Subtle elevation
- glow: Colored accent glow
- card, card-hover

2. Update src/app/globals.css with:

- CSS variables for light/dark themes
- .glass and .glass-dark utilities
- .gradient-text utility
- .acr-appropriate, .acr-maybe, .acr-inappropriate classes
- Animation delay utilities (.animation-delay-100, etc.)
- .scrollbar-hide utility

3. Create src/lib/utils.ts with:

- cn() function using clsx + tailwind-merge
- formatCurrency(amount: number)
- formatRadiation(mSv: number)
- getRadiationCategory(mSv): 'none' | 'minimal' | 'low' | 'moderate' | 'high'
- getACRCategory(rating): 'usually-appropriate' | 'may-be-appropriate' | 'usually-not-appropriate'
- getACRColor(rating): { bg, text, border }
- calculateScore(acrRating): number (1-9 → 0-100)
- formatDuration(seconds)
- getTimeAgo(date)
- slugify(text)

Prompt 4: Core UI Components Part 1

Create production-quality UI components with TypeScript, accessibility, and Framer Motion.

1. src/components/ui/button.tsx

Using class-variance-authority (cva):

VARIANTS: primary, secondary, outline, ghost, danger, success

SIZES: xs, sm, md, lg, xl

FEATURES:

- Loading state with spinner
- Disabled state
- Icon support (left/right)
- Full-width option
- forwardRef
- Framer Motion tap animation (scale 0.98)
- Proper focus ring

2. src/components/ui/card.tsx

VARIANTS: default, elevated, outlined, interactive, glass

FEATURES:

- CardHeader, CardContent, CardFooter sub-components
- Padding options (none, sm, md, lg)
- Hover lift for interactive variant
- Optional accent border-left
- Compound component pattern

3. src/components/ui/input.tsx

FEATURES:

- Label with required indicator
- Error state with message
- Helper text
- Left/right icons
- Variants: default, filled
- Sizes: sm, md, lg
- Focus ring with accent
- forwardRef with proper aria attributes

4. src/components/ui/badge.tsx

VARIANTS: default, primary, success, warning, danger, info, em, im, fm, surgery, peds

SIZES: sm, md, lg

FEATURES:

- Optional dot indicator
- Optional icon
- Outline variant
- Pill shape

All components must support className override via cn() and include JSDoc comments.

Prompt 5: Core UI Components Part 2

Continue the UI library with interactive components using Radix UI primitives.

1. src/components/ui/modal.tsx

Using @radix-ui/react-dialog:

- Backdrop with fade + blur
- Content with scale-in animation
- Sizes: sm, md, lg, xl, full
- Close button, escape/backdrop close
- Focus trap, scroll lock

2. src/components/ui/tabs.tsx

Using @radix-ui/react-tabs:

- Variants: default (underline), pills, segmented
- Animated sliding indicator
- Content fade transition
- Keyboard navigation

3. src/components/ui/progress.tsx

LinearProgress:

- Animated fill
- Label support
- Size variants (sm, md, lg)

- Color variants (default, success, warning, danger)

CircularProgress:

- SVG-based circle
- Center label option

4. src/components/ui/select.tsx

Using @radix-ui/react-select:

- Styled trigger matching Input
- Animated dropdown
- Option groups
- Checkmark on selected
- Search/filter option

5. src/components/ui/checkbox.tsx

Using @radix-ui/react-checkbox:

- Animated checkmark (draw effect)
- Label, error, disabled states
- Indeterminate state

6. src/components/ui/switch.tsx

Using @radix-ui/react-switch:

- Smooth transition
- Size variants
- Label support

7. src/components/ui/tooltip.tsx

Using @radix-ui/react-tooltip:

- Positions: top, bottom, left, right
- Delay options
- Arrow pointer

8. src/components/ui/skeleton.tsx

VARIANTS: text, circle, rectangle, card

- Shimmer animation
- Customizable dimensions

Prompt 6: Layout Components

Create the navigation and layout shell components for ARKA-ED.

1. src/components/layout/navbar.tsx

Marketing nav:

- Logo: "ARKA" in accent, "-ED" in slate
- Links: Features, Pricing, About
- Auth: Log In, Get Started buttons
- Sticky with backdrop blur on scroll
- Mobile hamburger with slide-out drawer
- Framer Motion animations

2. src/components/layout/dashboard-sidebar.tsx

Structure:

- ARKA-ED Logo
- LEARN section: Cases, Specialty Tracks
- ASSESS section: Quizzes, Assessments
- TRACK section: Progress, Achievements
- Admin section (if admin role)
- User profile at bottom with dropdown

Features:

- Collapsible on desktop
- Drawer on mobile
- Active state indicator (left accent border + bg)
- Lucide icons for nav items

3. src/components/layout/dashboard-header.tsx

- Dynamic page title
- Breadcrumbs (optional)
- Search bar
- Notifications bell with badge
- User avatar dropdown (Profile, Settings, Logout)

4. src/components/layout/dashboard-shell.tsx

Combines sidebar + header + main content:

- Responsive (sidebar collapses mobile)

```
responsive (screen sizes, mobile,
```

- Consistent padding
- Page transition wrapper

```
## 5. src/components/layout/footer.tsx
```

Marketing footer:

- Logo + tagline
- Product, Resources, Company, Legal links
- Social icons
- Copyright

```
## 6. src/components/layout/admin-sidebar.tsx
```

Admin-specific with darker theme:

- Dashboard, Cases, Users, Analytics, Settings
- Back to App link

Prompt 7: Database Schema & Types

Create complete TypeScript types for ARKA-ED's database.

```
## src/types/database.ts
```

Define these interfaces:

```
### Profile
```

- id: string (UUID)
- email: string
- full_name: string
- avatar_url: string | null
- role: 'student' | 'resident' | 'attending' | 'admin'
- institution: string | null
- specialty_track: 'em' | 'im' | 'fm' | 'surgery' | 'peds' | null
- training_year: number | null
- streak_count: number
- last_activity_date: string | null
- onboarding_completed: boolean

- created_at, updated_at: string

Case

- id, slug, title, chief_complaint: string
- clinical_vignette: string (full case)
- patient_age: number
- patient_sex: 'male' | 'female'
- patient_history: string[]
- vital_signs: VitalSigns | null
- physical_exam: string | null
- lab_results: LabResult[] | null
- category: 'low-back-pain' | 'headache' | 'chest-pain' | 'abdominal-pain' | 'extremity-trauma'
- specialty_tags: ('em' | 'im' | 'fm' | 'surgery' | 'peds')[]
- difficulty: 'beginner' | 'intermediate' | 'advanced'
- acr_topic: string
- optimal_imaging: string[] (imaging_option IDs)
- explanation: string
- teaching_points: string[]
- clinical_pearls: ClinicalPearl[] | null
- hints: string[] | null
- references: Reference[]
- is_published: boolean
- timestamps

VitalSigns

heart_rate, blood_pressure_systolic/diastolic, respiratory_rate, temperature, temperature_unit, oxygen_saturation

LabResult

name, value, unit, reference_range, is_abnormal

ClinicalPearl

content, category: 'clinical-pearl' | 'high-yield' | 'common-mistake' | 'board-favorite'

Reference

title, source, year, url?

ImagingOption

id, name, short_name, modality, body_region, with_contrast, typical_cost_usd, radiation_msv, description, common_indications, contraindications, duration, is_active

CaseImagingRating

id, case_id, imaging_option_id, acr_rating (1-9), rating_category, rationale, acr_reference

UserCaseAttempt

id, user_id, case_id, selected_imaging, acr_rating_received, score, is_correct, time_spent_seconds, mode, assessment_id, hints_used, feedback_viewed, created_at

Assessment

id, title, description, case_ids, time_limit_minutes, passing_score, specialty_track, difficulty, is_published, is_custom, created_by, timestamps

UserAssessment

id, user_id, assessment_id, status, current_case_index, score, passed, started_at, completed_at, time_remaining_seconds, answers: AssessmentAnswer[]

Achievement

id, slug, name, description, icon, category, requirement_type, requirement_value, is_active

UserAchievement

id, user_id, achievement_id, achieved_at, metadata

UserProgress (computed)

user_id, total_cases_attempted/completed, overall_accuracy, current_streak, longest_streak, total_time_spent, category_progress[], specialty_progress[]

Export Database type for Supabase with Tables, Views, Functions, Enums.

Prompt 8: Supabase Migration SQL

Generate complete Supabase SQL migration.

Create supabase/migrations/001_initial_schema.sql

Enums:

user_role, specialty_track, case_category, difficulty_level, modality, acr_category, attempt_mode, assessment_status, achievement_category

Tables (with proper constraints):

1. profiles - extends auth.users
2. cases - with CHECK constraints, JSONB for vital_signs/lab_results
3. imaging_options - CHECK radiation_msv >= 0
4. case_imaging_ratings - UNIQUE(case_id, imaging_option_id), CHECK acr_rating 1-9
5. user_case_attempts - all foreign keys with ON DELETE CASCADE
6. assessments - case_ids as UUID[]
7. user_assessments - answers as JSONB
8. achievements - master list
9. user_achievements - UNIQUE(user_id, achievement_id)

Indexes:

- cases: category, specialty_tags (GIN), difficulty, published
- user_case_attempts: user_id, case_id, created_at, mode
- user_assessments: user_id, status

Functions:

- update_updated_at() - trigger function
- handle_new_user() - creates profile on signup
- calculate_user_progress(user_id) - returns progress stats

Triggers:

- updated_at on all tables
- on_auth_user_created ⇒ handle_new_user()

RLS Policies:

- profiles: own read/update
- cases: public read published, admin CRUD
- imaging_options: public read active, admin manage
- case_imaging_ratings: public read, admin manage
- user_case_attempts: own CRUD
- assessments: public read published, users create custom, admin manage
- user_assessments: own manage
- achievements: public read active
- user_achievements: own read, system insert

Phase 2: Marketing & Auth

Prompt 9: Landing Page Hero

Create a stunning landing page hero at `src/app/(marketing)/page.tsx`.

Design Direction:

- Deep navy background (#0f172a) with gradient mesh
- ARKA cyan accent (#06b6d4)
- Editorial typography
- Premium animations, not gimmicky
- Mobile-first responsive

HERO SECTION:

- Badge: "The Future of Medical Education"
- Headline: "Master Medical Imaging <gradient>Appropriateness</gradient>"
- Subheadline: "The first interactive platform teaching physicians when to order imaging — not just how to read it."
- StatsBar (animated count-up):
 - 97% of residents unfamiliar with ACR Criteria
 - \$100B+ wasted annually
 - 50+ clinical cases
- CTAs: "Start Learning Free" (primary), "Watch Demo" (secondary with play icon)
- Animated mockup of case interface

ANIMATIONS:

- Headline: Staggered word reveal (fade up, 100ms delay)
- Stats: Count-up when visible
- Background: Subtle gradient shift
- Floating elements: Gentle parallax on mouse

PROBLEM SECTION:

- Badge: "The Problem" (danger variant)
- Headline: "Every Platform Teaches the Wrong Skill"
- Comparison table:

- | What Others Teach | What You Need |
- | Reading CT scans | WHEN to order |
- | Identifying pathology | RIGHT modality |
- | Radiologist skills | Ordering physician skills |
- Statistics cards with sources:
 - 2.4% use ACR Criteria
 - 50%+ residents unfamiliar
 - \$100B wasted annually

Use Framer Motion for scroll-triggered animations and Intersection Observer.

Prompt 10: Landing Page Features & Pricing

Complete the landing page with remaining sections.

HOW IT WORKS SECTION:

ProcessFlow with 4 animated steps:

1. Read the Case (clipboard icon)
2. Order Imaging (mouse pointer icon)
3. Get Instant Feedback (checkmark icon)
4. Learn the Evidence (lightbulb icon)

Animated connectors, sequential reveal on scroll.

FEATURES GRID (3 columns):

1. 50+ Clinical Cases (BookOpen, cyan)
2. ACR Criteria Integration (Target, emerald)
3. Cost & Radiation Data (DollarSign, amber)
4. Specialty Tracks (Route, violet)
5. Progress Analytics (BarChart3, blue)
6. Assessment Mode (Award, rose)

Each card: icon, title, description, hover lift.

TESTIMONIALS SECTION:

- Logo bar (placeholder institutions)

- Testimonial carousel with quotes, avatars, institutions

PRICING SECTION:

Three tiers:

1. Free (\$0):

- 10 cases, basic progress, learning mode, community support
- CTA: "Get Started Free"

2. Pro (\$14.99/mo or \$99/yr) - FEATURED:

- All cases, full analytics, assessment mode, specialty tracks, achievements
- CTA: "Start 7-Day Free Trial"

3. Institutional (Custom):

- Everything + bulk licensing, LMS, admin dashboard, custom assessments
- CTA: "Contact Sales"

FINAL CTA:

- Gradient background (primary-900)
- Headline: "Ready to Order Smarter?"
- Email capture form
- Social proof: avatar stack + "500+ learners"

Prompt 11: Marketing Components

Create reusable marketing components in src/components/marketing/:

1. hero.tsx - Full viewport hero with gradient background, animations
2. feature-card.tsx - Icon, title, description, color, hover animation
3. stats-bar.tsx - Horizontal stats with count-up, icons, responsive
4. process-flow.tsx - Numbered steps, connecting lines, stagger animation
5. testimonial-card.tsx - Quote, author info, optional rating
6. pricing-card.tsx - Tier, price, features list, CTA, featured badge
7. comparison-table.tsx - Two-column right/wrong comparison
8. cta-section.tsx - Full-width CTA with email capture option
9. section-header.tsx - Badge, headline, subtitle pattern

All with Framer Motion, responsive design, className support.

Prompt 12: Authentication Pages

Create authentication pages with professional medical aesthetic.

src/app/(auth)/layout.tsx

Split layout:

- Left (50%): Form on white background
- Right (50%): Branded visual with quote, hidden on mobile

src/app/(auth)/login/page.tsx

- Email, password (show/hide toggle)
- Remember me checkbox
- Forgot password link
- Google OAuth button
- Link to register
- Form validation: react-hook-form + zod
- Loading state, error toast
- Redirect to /cases on success

src/app/(auth)/register/page.tsx

- Full name, email
- Password with strength indicator (bar + checklist)
- Confirm password
- Role dropdown (Student, Resident, Attending)
- Institution (optional)
- Terms checkbox
- Validation, loading, redirect to /onboarding

src/app/(auth)/forgot-password/page.tsx

- Email input
- Submit → "Check your email" success state
- Back to login link

Components:

- auth-form.tsx - Consistent wrapper
- social-buttons.tsx - Google, Microsoft OAuth

- Password strength indicator

```
## src/lib/validations.ts
```

```
Export zod schemas: loginSchema, registerSchema, forgotPasswordSchema
```

Prompt 13: Auth Middleware & Guards

Implement comprehensive route protection.

```
## src/middleware.ts
```

Complete implementation:

- PUBLIC_ROUTES: /, /login, /register, /forgot-password, /pricing, /about
- AUTH_ROUTES: /login, /register, /forgot-password (redirect if authed)
- PROTECTED_ROUTES: /cases/*, /progress/*, /assessments/*, etc.
- ADMIN_ROUTES: /admin/* (check profile.role === 'admin')

Logic:

- Create Supabase client
- Get user
- Check route type
- Redirect unauthenticated from protected → /login?redirect=path
- Redirect authenticated from auth → /cases
- Check admin role for admin routes

```
## src/lib/auth/get-user.ts
```

Server-side helpers:

- getUser() - returns user or null
- getProfile() - returns profile with role
- requireAuth() - redirects if not authenticated
- requireAdmin() - redirects if not admin

```
## src/components/auth/auth-guard.tsx
```

Client wrapper:

- Loading state
- Check auth on mount
- Listen to onAuthStateChange

- Redirect if not authenticated

src/components/auth/role-guard.tsx

- allowedRoles prop
- Check profile.role
- Redirect or show fallback

src/lib/hooks/use-auth.ts

Client hook:

- user, profile, loading state
- signIn, signUp, signOut, signInWithGoogle
- isAdmin computed

Prompt 14: Onboarding Flow

Create welcoming onboarding at src/app/(auth)/onboarding/page.tsx.

Multi-step wizard (4 steps):

Step 1: Welcome

- Animated illustration
- "Welcome to ARKA-ED, {firstName}!"
- "Let's personalize your experience"
- "Let's Get Started" button
- Skip link

Step 2: Role & Experience

- Role cards (Medical Student, Resident, Attending)
- If student: Year selector (MS1-MS4)
- If resident: Training year selector (PGY-1 through Fellow)

Step 3: Specialty Interest

- Specialty cards with colors:
 - EM (red, siren icon)
 - IM (blue, heart icon)
 - FM (green, users icon)

- PVI (green, users icon)
- Surgery (purple, scissors icon)
- Peds (teal, baby icon)
- Case count per specialty

Step 4: Quick Tutorial

- Carousel with 3 slides:
 1. Read the case (mockup)
 2. Choose imaging (mockup)
 3. Get feedback (mockup)
- "Start My First Case" button

Implementation:

- State management with useState or Zustand
- Save progress to DB each step
- Framer Motion step transitions
- Progress indicator (dots/bar)
- On completion: profile.onboarding_completed = true
- Redirect to /cases

Phase 3: Core Learning Experience

Prompt 15: Case Library Page

Create the main case library at src/app/(dashboard)/cases/page.tsx.

Page Structure:

Header:

- "Case Library" title
- "Practice with real clinical scenarios"

Progress Summary (animated):

- X/50 cases completed
- Y% accuracy
- 🔥 Z day streak

Filters Bar:

- Category: All, Low Back Pain, Headache, Chest Pain, Abdominal Pain, Extremity Trauma
- Specialty: All, EM, IM, FM, Surgery, Peds
- Difficulty: All, Beginner, Intermediate, Advanced
- Status: All, Not Started, In Progress, Completed
- Search input (debounced)
- Sort: Alphabetical, Newest, Difficulty

Case Grid:

- 3 cols desktop, 2 tablet, 1 mobile
- CaseCard components
- Staggered entrance animation
- Empty state if no results

Pagination:

- 12 per page
- Page numbers, prev/next

Data Fetching (Server Component):

- Query cases with user_case_attempts join
- Apply filters from searchParams
- Use URL params for state (shareable)
- Call calculate_user_progress RPC

Components:

- ProgressSummary
- CaseFilters (uses URL search params)
- CaseGrid
- Pagination

Prompt 16: Case Card Component

Create CaseCard at src/components/cases/case-card.tsx.

Visual Structure:

- Category icon + badge
- Difficulty badge (color coded)
- Title (2 lines max)
- Chief complaint quote (2 lines)
- Specialty tags
- Time estimate (~5 min)
- Status: Not Started / In Progress / Completed with score

Implementation:

- Accept case data with user_attempts
- Determine status from latest attempt
- Category configs (icon, label)
- Difficulty configs (color, label)
- Status rendering:
 - Completed: Green check, score badge
 - In Progress: Yellow rotate icon, "Try Again"
 - Not Started: Gray circle
- Link to /cases/[slug]
- Motion animation (fade up, stagger by index)
- Hover lift effect
- Border-left accent for completed

Prompt 17: Individual Case Page

Create case page at src/app/(dashboard)/cases/[caseId]/page.tsx.

Desktop Layout (split view):

Left Panel (60%): Clinical Case

Right Panel (40%): Ordering Interface ⇒ Feedback Panel after submit

Data Fetching:

- Query case with case_imaging_ratings → imaging_options
- Get user's previous attempts
- Redirect to /cases if not found

CaseViewer Component (src/components/cases/case-viewer.tsx):

```
## Case Header Component (src/components/cases/case-header.tsx):
```

State:

- selectedImaging: string[]
- mode: 'learning' | 'quiz'
- isSubmitted: boolean
- feedbackData

Structure:

- CaseHeader (title, mode toggle, timer if quiz)
- Split panels
- Left: ClinicalVignette
- Right: OrderingInterface OR FeedbackPanel

ClinicalVignette Component:

- PatientCard (age, sex, chief complaint with avatar)
- HPI section
- Past Medical History (if any)
- Vital Signs (formatted)
- Physical Exam (collapsible)
- Lab Results (table, collapsible)
- Clinical Question card
- Hints (learning mode only)

PatientCard:

- Avatar circle with icon
- "[age]-year-old [sex]"
- Chief Complaint: "[quote]"

Mobile Layout:

- Tabs: "Case" | "Order"
- Sticky submit at bottom

Prompt 18: Ordering Interface

Create OrderingInterface at src/components/cases/ordering-interface.tsx.

Visual Structure:

- Search bar
- Modality tabs: All | X-ray | CT | MRI | US | Nuclear | None
- Scrollable imaging options list
- Divider with "or"
- "No Imaging Indicated" special option
- Selection summary (count, total cost, radiation)
- Submit button

Props:

- `imagingOptions: ImagingOption[]`
- `selectedImaging: string[]`
- `onSelectionChange: (ids) => void`
- `onSubmit: () => void`
- `mode: 'learning' | 'quiz'`
- `disabled?: boolean`

Features:

- Search filter (debounced)
- Modality tab filter
- Toggle selection:
 - If selecting "No Imaging", clear others
 - If selecting imaging, remove "No Imaging"
- Calculate totals (cost, radiation)
- `AnimatePresence` for filtered list

ImagingOptionCard Component:

- Checkbox with animated check
- Name
- Modality icon (color coded)
- Cost: \$XXX
- Radiation: X.X mSv with visual bar
 - None (green): 0
 - Minimal: <0.1
 - Low: 0.1-1
 - Moderate: 1-10
 - High: >10
- Selected state (scale + border)
- "special" variant for No Imaging

Selection Summary:

- "X selected" or "No selection"
- Total cost and radiation
- Clear button

Prompt 19: Feedback Panel

Create FeedbackPanel at src/components/cases/feedback-panel.tsx.

CRITICAL for learning - make it rewarding and educational.

Structure:**### Result Header:**

- Large icon (✓ green / ✗ red / ~ yellow)
- Message: "Excellent!" / "Good Thinking" / "Let's Review"
- Animated score: XX/100
- Confetti for score ≥ 90

Your Selection Card:

- Selected imaging name
- ACR Rating Badge (1-9 with color)
- Rationale text
- Cost and radiation

Better Choice Card (if not correct):

- Optimal imaging
- ACR Rating (7-9)
- Why it's better

ACR Scale Visualization:

- Horizontal 1-9 scale
- Gradient (red $\Rightarrow$ yellow $\Rightarrow$ green)
- User choice marker

Optimal marker (if different)

- Optimal marker (if different)

Explanation:

- "Why?" heading with book icon
- Detailed explanation text

Teaching Points:

- Numbered list (1-4 items)
- Lightbulb icon
- Memorable facts

All Options Comparison (Learning Mode):

- Collapsible accordion
- Table: Option | ACR | Cost | Radiation
- Sorted by ACR (best first)
- Highlight user's selection

References:

- Collapsible
- Title, source, year, link

Action Buttons:

- "Next Case" (primary)
- Review Case (rotate icon)
- Save/Bookmark (bookmark icon)

Animations:

- Panel slides in from right
- Score counter animates
- Staggered section reveal

Prompt 20: ACR Rating Components

Create specialized ACR rating display components.

1. src/components/cases/acr-rating-badge.tsx

Props: rating (1-9), size ('sm' | 'md' | 'lg'), showLabel

- Circular badge with number
- Colors:
 - 1-3: Rose bg, dark text
 - 4-6: Amber bg, dark text
 - 7-9: Emerald bg, dark text
- Label below: "Usually Not/May Be/Usually Appropriate"
- Tooltip explaining scale

2. src/components/cases/acr-rating-scale.tsx

Props: userRating?, optimalRating?

- Horizontal bar 1-9
- Gradient background (red → yellow → green)
- Section labels
- Animated markers for ratings
- Legend

3. src/components/cases/radiation-indicator.tsx

Props: mSv, showLabel?, size?

- Radiation symbol icon
- Visual bar/meter
- Color coding by category
- Reference text ("equivalent to X chest X-rays")
- Tooltip with details

4. src/components/cases/cost-display.tsx

Props: costUSD, comparison?

- Formatted currency
- Comparison arrow if provided (↑ red, ↓ green)

Prompt 21: Learning Mode Engine

Implement learning mode features.

1. src/components/cases/learning-mode-toggle.tsx

- Toggle switch: Learning ↔ Quiz

- Learning: Hints available, no timer, detailed feedback
- Quiz: No hints, timer, score tracking
- Animated toggle with icons
- Persist preference to localStorage

2. src/components/cases/hint-system.tsx

Props: hints: string[], maxHints: number

- "Need a hint?" button
- Progressive reveal (one at a time)
- First hint free, subsequent cost points
- Callout box for hint display
- Track hints used for analytics

3. src/components/cases/clinical-pearl.tsx

Props: pearl: string, category

- Highlighted callout box
- Lightbulb icon
- Categories: "Clinical Pearl", "High-Yield", "Common Mistake", "Board Favorite"
- Distinct styling per category

4. Update case page for Learning Mode:

- Highlighted keywords in vignette (hover for definitions)
- "Why might this be important?" expandable sections
- Related cases suggestions

5. src/lib/hooks/use-learning-mode.ts

- mode state
- hint tracking
- time spent
- auto-save
- localStorage persistence

Prompt 22: Quiz Mode & Timer

Implement quiz mode with timer.

Update CaseViewer for Quiz Mode:

- Timer display in header
- No hints available
- Score tracking
- Auto-submit when time expires

src/components/cases/quiz-timer.tsx

Props: initialSeconds, onExpire, isPaused

- Countdown display (MM:SS)
- Warning states:
 - Normal: Default color
 - 5 min left: Yellow
 - 1 min left: Red, pulsing
- Pause/resume (if allowed)
- Progress ring animation
- Auto-submit callback on expire

Quiz Mode Logic:

- Start timer on case load
- Record time spent per case
- No feedback until submission
- Score based on ACR rating
- Store attempt with mode: 'quiz'

Phase 4: Assessment System

Prompt 23: Assessment List Page

Create assessment list at src/app/(dashboard)/assessments/page.tsx.

Page Structure:

- "Assessments" header
- "Test your knowledge with comprehensive evaluations"
- Filters: Specialty, Difficulty, Status
- "Create Custom Quiz" button

Assessment Cards (src/components/assessments/assessment-card.tsx):

- Title & description
- Case count
- Time limit (if any)
- Passing score
- Specialty badge
- Difficulty badge
- User status:
 - Not Started: Gray, "Start" button
 - In Progress: Yellow, "Continue" button, progress %
 - Completed: Green, best score, "Retake" option

Data Fetching:

- Query assessments with user_assessments join
- Filter by is_published
- Include custom (created_by = user)

Prompt 24: Assessment Taking Flow

Create assessment flow at src/app/(dashboard)/assessments/[assessmentId]/page.tsx.

States:

Start Screen:

- Title, description
- Instructions card
- Stats: X cases, Y minutes, Z% to pass
- "Begin Assessment" button

During Assessment:

- Progress bar (Case X of Y)
- Timer (if timed)
- Case content (same as individual case)
- Submit → next case (no immediate feedback)
- "Flag for Review" toggle

Navigation: can review flagged cases

- Navigation: Can review flagged cases

Completion Screen:

- Overall score with animation
- Pass/Fail badge
- Performance breakdown:
 - By category
 - By difficulty
- Percentile comparison
- Review incorrect answers option
- Certificate/badge if passed
- "Retake" / "Back to Assessments" buttons

src/components/assessments/quiz-progress.tsx:

- Horizontal progress bar
- Case markers (answered, flagged, current)
- Click to navigate (if allowed)

src/components/assessments/results-breakdown.tsx:

- Category scores (bar charts)
- Difficulty scores
- Time per case
- Weak areas identification

Prompt 25: Assessment Results

Create results page at src/app/(dashboard)/assessments/[assessmentId]/results/page.tsx.

Structure:

Score Hero:

- Large animated score
- Pass/Fail with confetti if passed
- Percentile rank

Performance Analysis:

- Overall stats card
- Category breakdown (horizontal bars)
- Difficulty breakdown
- Time analysis (average per case, total)

Missed Questions Review:

- List of incorrect cases
- For each:
 - Case title
 - Your answer vs correct
 - ACR ratings
 - "Review Case" link

Improvement Suggestions:

- Weak areas identified
- Recommended cases to practice
- Related assessments

Actions:

- "Retake Assessment"
- "Practice Weak Areas"
- "Back to Assessments"
- Share results (copy link)

Prompt 26: Custom Quiz Builder

Create quiz builder at `src/app/(dashboard)/assessments/create/page.tsx`.

Multi-step wizard:

Step 1: Select Topics

- Checkbox grid of categories
- Select all / deselect all
- Case count per category
- Visual preview of coverage

Step 2: Set Parameters

- Number of cases: Slider 5-30
- Difficulty mix: Three sliders (beginner/intermediate/advanced %)
- Time limit: Toggle + input (minutes)
- Randomize order: Toggle

Step 3: Specialty Focus (optional)

- Specialty selector
- Or "All Specialties"

Step 4: Review & Start

- Summary of selections
- Estimated time
- "Start Quiz" button

src/components/assessments/quiz-builder.tsx:

- Multi-step form
- Animated transitions
- Progress indicator
- Back/Next navigation
- Form state with react-hook-form

Features:

- Save quiz template
- "My Saved Quizzes" section
- Share configuration (URL or code)

Phase 5: Progress & Gamification

Prompt 27: Progress Dashboard

Create progress page at src/app/(dashboard)/progress/page.tsx.

Structure:

Header:

- "Your Progress"
- Date range: This Week | This Month | All Time
- Streak counter with fire emoji

Stats Overview (4 cards):

1. Cases Completed: X / Total (with ring chart)
2. Accuracy Rate: X% (trend arrow)
3. Current Streak: X days (fire animation)
4. Time Invested: X hours

Performance Over Time:

- Line chart (Recharts)
- X: dates, Y: accuracy %
- Tooltip with details
- Animated on mount

Competency Radar:

- Spider/radar chart
- 5 axes: Low Back Pain, Headache, Chest Pain, Abdominal, Trauma
- Current level vs target
- Animated drawing

Category Breakdown:

- Horizontal bars per category
- Attempted, Correct, Accuracy %
- Color by performance
- Click to filter cases

Recent Activity:

- Timeline feed
- Date, case title, score, time
- Quick link to review
- "Load More" pagination

Achievements:

- Grid of badges
- Earned: colored, animated
- Locked: grayscale

- Click for details

Weak Areas:

- Lowest categories

- Suggested cases

- "Focus Mode" button

Prompt 28: Charts & Visualizations

Create chart components in src/components/progress/:

1. progress-chart.tsx

Recharts LineChart wrapper:

- Props: data, dateRange, metric
- Responsive
- Custom tooltip
- Animated on mount

2. competency-radar.tsx

Recharts RadarChart wrapper:

- Props: categories, currentScores, targetScores
- Animated drawing
- Legend
- Hover details

3. category-bar.tsx

Horizontal progress bars:

- Props: category, attempted, correct, total
- Animated fill
- Percentage label
- Color by performance

4. streak-calendar.tsx

GitHub-style contribution calendar:

- Shows daily activity

- Color intensity by cases completed
- Tooltip with date and count
- Current streak highlight

5. stats-card.tsx

Reusable stat display:

- Props: title, value, trend, icon
- Trend arrow (up green, down red)
- Animated counter

6. activity-feed.tsx

Vertical timeline:

- Props: activities[]
- Relative time
- Score color coding
- Link to case review

Prompt 29: Achievement System

Create achievement system.

src/data/achievements.ts

Define achievements:

Progress:

- first-case: Complete first case
- ten-cases: Complete 10 cases
- fifty-cases: Complete 50 cases
- hundred-cases: Complete 100 cases

Mastery:

- perfect-score: Get 100% on a case
- category-expert: 90%+ in a category
- all-categories: Complete case in every category
- speedster: Complete case in under 2 minutes

Streak:

- three-day-streak: 3 days in a row
- week-streak: 7 days
- month-streak: 30 days

Special:

- early-adopter: Join during beta
- feedback-hero: Submit 5 feedback reports
- quiz-master: Pass assessment with 100%

src/components/progress/achievement-badge.tsx

Props: achievement, isEarned, date

- Circular badge with icon
- Grayscale if not earned
- Shine animation when earned
- Modal on click with details

Achievement Modal:

- Large badge display
- Name and description
- Date earned
- Progress if not earned
- Share button

Achievement Checking Logic:

- After each case completion, check criteria
- Unlock new achievements
- Show toast notification
- Trigger confetti for special ones

Prompt 30: Streak & Gamification

Implement streak system and gamification.

Streak Logic:

- Check last_activity_date on each visit

- If same day: continue streak
- If yesterday: increment streak
- If older: reset streak to 1

src/components/progress/streak-display.tsx

- Fire emoji with number
- Animated flame for active streak
- "X day streak!" message
- Longest streak comparison
- Streak freeze feature (future)

Streak Notifications:

- Toast on streak milestone (3, 7, 14, 30 days)
- Achievement unlock
- Confetti effect

Leaderboard (Optional):

- Weekly/monthly rankings
- Anonymous or opted-in
- Case count, accuracy, streak

Points System (Optional):

- Points per case: ACR rating × multiplier
- Bonus: streak, speed, hints unused
- Display on progress page

Daily Goals:

- "Complete 3 cases today"
- Progress bar
- Reward: streak protection, bonus points

Phase 6: Specialty Tracks

Prompt 31: Specialty Track Pages

Create specialty tracks at `src/app/(dashboard)/specialty/[track]/page.tsx`.

Dynamic route for: em, im, fm, surgery, peds

Track Configurations:

- EM: Red theme, siren icon
- IM: Blue theme, heart icon
- FM: Green theme, users icon
- Surgery: Purple theme, scissors icon
- Peds: Teal theme, baby icon

Page Structure:

Track Header:

- Large icon with color
- "[Specialty] Track" title
- Description
- Progress: X/Y cases, Z% complete
- Track selector dropdown

Curriculum:

- Organized into modules/units
- Example for EM:
 - Unit 1: Trauma Imaging
 - Unit 2: Chest Pain Evaluation
 - Unit 3: Abdominal Emergencies
 - Unit 4: Neurological Emergencies
 - Unit 5: Pediatric EM Cases

Module Cards:

- Title and description
- Case count
- Completion progress bar
- Locked/unlocked state (sequential)

Case List:

- Filtered for specialty
- Sorted by curriculum order
- Progress indicators

Track Assessment:

- Specialty-specific exam
- "Test Your Knowledge" CTA
- Requirement: Complete X cases first
- Certificate on pass

Resources:

- Related ACR topics
- Recommended reading
- Clinical guidelines

Prompt 32: Curriculum Module System

Create curriculum components.

src/components/specialty/track-header.tsx

- Track icon and name with color
- Progress overview
- Track selector dropdown
- Description

src/components/specialty/curriculum-module.tsx

Props: module, progress, locked

- Expandable card
- Title, description
- Case list within
- Sequential unlock badge
- Completion celebration

src/components/specialty/track-progress.tsx

- Overall progress ring
- Module completion checklist
- Estimated time remaining

src/data/specialty-tracks.ts

Define for each specialty:

- id, name, slug, description
- icon, color scheme
- curriculum: modules with case IDs
- prerequisites
- assessment requirements

Example:

```
``typescript
export const specialtyTracks = {
  em: {
    id: 'em',
    name: 'Emergency Medicine',
    slug: 'em',
    color: 'red',
    icon: 'Siren',
    description: 'Master imaging decisions in the acute care setting',
    curriculum: [
      {
        id: 'em-trauma',
        title: 'Trauma Imaging',
        description: 'Appropriate imaging in trauma scenarios',
        caseIds: ['trauma-head', 'trauma-cspine', ...],
        unlockRequirement: null,
      },
      // ...
    ],
    assessmentId: 'em-comprehensive',
    certificateName: 'Emergency Imaging Appropriateness Certificate',
  },
  // ...other specialties
}
```

Phase 7: Admin Panel

Prompt 33: Admin Dashboard

Create admin dashboard at `src/app/admin/page.tsx`.

Access Control:

- RoleGuard for admin only
- Distinct dark theme

Dashboard Stats:

- Total users (with growth %)
- Active users (daily/weekly/monthly)
- Total case attempts
- Average accuracy
- Popular cases (top 5)

Quick Actions:

- Add New Case
- View Users
- Generate Report
- Manage Assessments

Recent Activity Feed:

- New registrations
- Case completions
- Assessment passes
- Error reports

System Health:

- Database status
- Error rate
- Performance metrics

Prompt 34: Case Management

Create case management at `src/app/admin/cases/page.tsx`.

Table View:

Columns:

- Title
- Category (badge)
- Difficulty (badge)
- Status (Published/Draft)
- Attempts count
- Avg score
- Created/Updated
- Actions (Edit, Duplicate, Archive, Delete)

Features:

- Search and filter
- Bulk actions (publish, unpublish, delete)
- Sorting by any column
- Pagination
- "Add New Case" button

src/components/admin/case-table.tsx

- Server-side pagination
- Column sorting
- Row selection
- Action dropdowns
- Confirmation modals for destructive actions

Prompt 35: Case Editor

Create case editor at `src/app/admin/cases/[caseId]/edit/page.tsx` and `src/components/admin/case-editor.tsx`.

Form Sections:

Basic Info:

- Title (required)
- Slug (auto-generated, editable)
- Category dropdown
- Difficulty dropdown

- Difficulty dropdown
- Specialty tags (multi-select)
- ACR Topic reference

Patient Info:

- Age (number)
- Sex (dropdown)
- Chief complaint

Clinical Vignette:

- History of Present Illness (rich textarea)
- Past Medical History (tag input)
- Medications (tag input)
- Social History
- Family History
- Review of Systems

Physical Exam & Vitals:

- Vital signs inputs (HR, BP, RR, Temp, SpO2)
- Physical exam findings

Lab Results:

- Dynamic add/remove
- Name, value, unit, reference, abnormal flag

Imaging Options:

- Select from master list
- For each, set:
 - ACR Rating (1-9)
 - Rationale
 - Is optimal? (checkbox)

Explanation & Teaching:

- Detailed explanation (rich text)
- Teaching points (list)
- Clinical pearls (list with category)
- Hints (list)

References:

- Add/remove
- Title, source, year, URL

Preview:

- Live preview tab
- Student view simulation

Features:

- Form validation with zod
- Auto-save draft
- Version history
- Publish/Unpublish toggle
- Duplicate case function

Prompt 36: User Management & Analytics

Create user management at `src/app/admin/users/page.tsx`.

User Table:

Columns:

- Name, Email
- Role (badge)
- Institution
- Cases completed
- Accuracy
- Last active
- Actions (View, Edit Role, Disable)

Features:

- Search by name/email
- Filter by role, institution
- Export CSV
- Invite users

User Detail Modal:

- Profile info
- Activity history
- Progress stats
- Role management

Analytics Page (src/app/admin/analytics/page.tsx):

Usage Trends:

- Daily/weekly/monthly active users
- Case attempts over time
- Registration trends

Content Analytics:

- Most attempted cases
- Hardest cases (lowest accuracy)
- Easiest cases
- Incomplete cases (started but not finished)

Performance Analytics:

- Score distribution histogram
- Average score by category
- Average score by difficulty
- Time spent analysis

Engagement:

- Retention cohorts
- Streak distribution
- Assessment completion rates

Phase 8: Content & Data

Prompt 37: Imaging Options Database

Create imaging options data at src/data/imaging-options.ts.

Structure:

```
``typescript
export const imagingOptions: ImagingOption[] = [
  // X-RAY
  {
    id: 'xray-chest',
    name: 'Chest X-ray (PA and Lateral)',
    shortName: 'CXR',
    modality: 'xray',

    bodyRegion: 'chest',
    withContrast: false,
    typicalCostUsd: 100,
    radiationMsv: 0.1,
    description: 'Standard chest radiograph...',
    commonIndications: ['Chest pain', 'Shortness of breath', ...],
    contraindications: ['Pregnancy (relative)'],
    duration: '5-10 minutes',
    isActive: true,
  },
  // ... more X-ray options

  // CT
  {
    id: 'ct-head-nc',
    name: 'CT Head without Contrast',
    shortName: 'CT Head',
    modality: 'ct',
    bodyRegion: 'head',
    withContrast: false,
    typicalCostUsd: 500,
    radiationMsv: 2.0,
    // ...
  },
  // ... more CT options

  // MRI
  {
    id: 'mri-brain-ne',
    name: 'MRI Brain without Contrast',
```

```

    shortName: 'MRI Brain',
    modality: 'mri',
    bodyRegion: 'brain',
    withContrast: false,
    typicalCostUsd: 1500,
    radiationMsv: 0,
    // ...
  },

  // ... more MRI options

  // ULTRASOUND
  // NUCLEAR

  // SPECIAL
  {
    id: 'no-imaging',
    name: 'No Imaging Indicated',
    shortName: 'None',
    modality: 'none',
    bodyRegion: 'none',
    withContrast: false,
    typicalCostUsd: 0,
    radiationMsv: 0,
    description: 'Conservative management without imaging',
    commonIndications: ['Self-limited conditions', 'Red flags absent'],
    contraindications: [],
    duration: 'N/A',
    isActive: true,
  },
]
'''

```

Include at minimum:

- 6 X-ray options (chest, lumbar, cervical, ankle, hip, abdomen)
- 8 CT options (head ±contrast, cervical, chest ±contrast, abd/pelvis ±contrast, CTA)
- 6 MRI options (brain ±contrast, lumbar ±contrast, cervical, knee)
- 4 Ultrasound options (abdomen, pelvis, LE venous, echo)
- 3 Nuclear options (VQ, bone scan, myocardial perfusion)

- 1 "No Imaging" option

Helper functions:

- `getImagingByModality(modality)`
- `getImagingByBodyRegion(region)`
- `calculateTotalCost(ids[])`
- `calculateTotalRadiation(ids[])`
- `getRadiationCategory(mSv)`

Prompt 38: Seed Cases — Low Back Pain

Create seed cases for Low Back Pain at `src/data/cases/low-back-pain.ts`.

Case 1: Acute Mechanical LBP (Beginner)

- Title: "Acute Low Back Pain in a Young Adult"
- Chief Complaint: "Low back pain for 3 days"
- Patient: 28 M
- Vignette: Acute onset after helping friend move furniture. Pain localized to lumbar region, worse with movement, relieved by rest. No radiation to legs, no numbness/tingling, no bowel/bladder changes. No fever, weight loss, cancer history.
- Vitals: Normal
- Exam: Tenderness L4-L5 paraspinal muscles, no neuro deficits
- Optimal: No Imaging (ACR 9)
- Teaching: Acute mechanical LBP < 6 weeks without red flags does not require imaging

Case 2: LBP with Red Flags (Intermediate)

- Title: "Low Back Pain with Neurological Symptoms"
- Chief Complaint: "Back pain with leg weakness"
- Patient: 55 F
- Vignette: 2 weeks progressive LBP, new left leg weakness, urinary retention. History of breast cancer 5 years ago.
- Optimal: MRI Lumbar Spine without contrast (ACR 9) - emergent
- Teaching: Red flags (cancer history + neuro symptoms) warrant urgent imaging

Case 3: Chronic LBP (Intermediate)

- Title: "Chronic Low Back Pain with Failed Conservative Therapy"
- Patient: 45 M

- Patient: 75 M

- Vignette: 4 months LBP, PT completed, NSAIDs tried, no improvement. No red flags.
- Optimal: MRI Lumbar Spine without contrast (ACR 7-8)
- Teaching: Chronic LBP > 6 weeks with failed conservative therapy may benefit from imaging

Case 4: LBP with Radiculopathy (Advanced)

- Title: "Low Back Pain with Sciatica"
- Patient: 38 M
- Vignette: 3 weeks LBP with shooting pain down right leg to foot. Positive straight leg raise. No red flags.
- Optimal: Start conservative. MRI if persistent > 6 weeks.
- Teaching: Radiculopathy without red flags: conservative first, image if no improvement

Case 5: Suspected Compression Fracture (Intermediate)

- Title: "Low Back Pain After Fall in Elderly Patient"
- Patient: 72 F
- Vignette: Fell at home 1 week ago, severe midline pain T12-L1. Osteoporosis. Pain on palpation.
- Optimal: X-ray Lumbar/Thoracic (ACR 8) or MRI if fracture confirmed
- Teaching: Elderly + fall + point tenderness = imaging indicated

For each case provide:

- Full vignette (100-200 words)
- Vital signs
- Physical exam
- All imaging options with ACR ratings (1-9) and rationale
- Detailed explanation
- 3-4 teaching points
- References

Prompt 39: Seed Cases — Headache & Chest Pain

Create seed cases for Headache and Chest Pain.

HEADACHE CASES:

Case 1: Chronic Daily Headache (Beginner)

- 32 F, bilateral pressure headaches daily x3 months

- No aura, no nausea, normal neuro exam
- Takes ibuprofen frequently
- Optimal: No Imaging (ACR 8-9)
- Teaching: Chronic stable headache without red flags = likely medication overuse

Case 2: Thunderclap Headache (Advanced)

- 45 M, sudden severe headache during exercise
- "Worst headache of my life"
- Neck stiffness, no fever
- Optimal: CT Head without contrast (ACR 9), then LP if negative
- Teaching: Thunderclap = SAH until proven otherwise

Case 3: New Headache with Papilledema (Intermediate)

- 28 F, obese, new progressive headaches
- Visual changes, papilledema on exam
- Optimal: MRI Brain with contrast (ACR 8)
- Teaching: Papilledema = elevated ICP, need imaging

Case 4: Migraine without Red Flags (Beginner)

- 25 F, recurrent unilateral headaches with aura
- Family history of migraine
- Normal exam
- Optimal: No Imaging (ACR 9)
- Teaching: Typical migraine with normal exam = clinical diagnosis

CHEST PAIN CASES:

Case 1: Typical Angina (Intermediate)

- 62 M, substernal pressure with exertion x2 months
- DM, HTN, HLD, non-smoker
- Optimal: Stress testing (ACR 8)
- Teaching: Intermediate pretest probability = functional testing

Case 2: Low-Risk Chest Pain (Beginner)

- 25 F, sharp left chest pain, worse with inspiration
- Reproducible on palpation
- Normal ECG
- Optimal: No Imaging (ACR 9) - likely musculoskeletal
- Teaching: Reproducible chest wall tenderness = likely MSK

Case 3: PE Suspected (Intermediate)

- 35 F, sudden pleuritic chest pain + dyspnea
- Recent long flight, on OCPs
- Wells score intermediate
- Optimal: CT Angiography Chest (ACR 8)
- Teaching: Intermediate Wells + positive D-dimer = CTPA

Case 4: Aortic Dissection Concern (Advanced)

- 58 M, sudden tearing chest pain radiating to back
- HTN, unequal arm BPs
- Optimal: CT Angiography Chest (ACR 9) - emergent
- Teaching: Classic dissection presentation = immediate CTA

Same format as LBP cases.

Prompt 40: Seed Cases — Abdominal & Trauma

Create seed cases for Abdominal Pain and Extremity Trauma.

ABDOMINAL PAIN CASES:

Case 1: RLQ Pain - Appendicitis (Intermediate)

- 22 M, periumbilical pain migrated to RLQ
- Fever, anorexia, positive McBurney's
- Optimal: CT Abdomen/Pelvis with contrast (ACR 8)
- Teaching: Classic appendicitis presentation = CT to confirm

Case 2: RUQ Pain - Biliary Colic (Intermediate)

- 45 F, postprandial RUQ pain
- Obese, Murphy's sign equivocal
- Optimal: Ultrasound RUQ (ACR 9)
- Teaching: Suspected biliary = US first (no radiation, excellent for gallstones)

Case 3: Epigastric Pain - Pancreatitis (Intermediate)

- 55 M, severe epigastric pain radiating to back
- Heavy alcohol use, elevated lipase
- Optimal: CT Abdomen/Pelvis with contrast (ACR 7-8) if diagnosis uncertain
- Teaching: Clinical + labs often sufficient; CT for complications

Case 4: Diffuse Abdominal Pain - Obstruction (Advanced)

- 68 M, prior abdominal surgeries
- Distension, vomiting, no BM x3 days
- Optimal: CT Abdomen/Pelvis with contrast (ACR 9)
- Teaching: SBO with prior surgery = CT to evaluate

EXTREMITY TRAUMA CASES:

Case 1: Ankle Injury - Ottawa Negative (Beginner)

- 30 M, twisted ankle playing basketball
- Weight-bearing, no bone tenderness at malleoli or proximal 5th MT
- Optimal: No Imaging (ACR 9) per Ottawa Ankle Rules
- Teaching: Ottawa Rules = validated clinical decision rule

Case 2: Ankle Injury - Ottawa Positive (Beginner)

- 28 F, ankle inversion injury
- Cannot bear weight, tender at lateral malleolus
- Optimal: X-ray Ankle (ACR 9)
- Teaching: Meets Ottawa criteria = X-ray indicated

Case 3: Elderly Fall - Hip Fracture (Intermediate)

- 78 F, fell at home, cannot bear weight
- Shortened and externally rotated leg
- Optimal: X-ray Hip (ACR 9), MRI if negative but high suspicion
- Teaching: Elderly + fall + hip pain = assume fracture until proven otherwise

Case 4: Knee Trauma - Possible Fracture (Intermediate)

- 35 M, direct blow to knee in MVA
- Swelling, unable to flex past 90°
- Optimal: X-ray Knee (ACR 8-9), Ottawa Knee Rules
- Teaching: Apply Ottawa Knee Rules for decision

Same detailed format.

Phase 9: Polish & Production

Prompt 41: Animations & Micro-interactions

Add polish throughout ARKA-ED with Framer Motion.

```
## src/components/animations/page-transition.tsx
```

- Wrap page content
- Fade + slide up on enter
- Fade out on exit
- AnimatePresence

```
## src/components/animations/fade-in.tsx
```

- Props: delay, duration, direction
- Intersection Observer trigger option

```
## src/components/animations/stagger-children.tsx
```

- Stagger children animations
- Props: staggerDelay, baseDelay

```
## Update existing components:
```

```
### Buttons:
```

- Scale down on press (0.98)
- Shadow change on hover
- Loading spinner rotation

```
### Cards:
```

- Lift on hover (translateY -2px, shadow increase)
- Border color change

```
### Modals:
```

- Backdrop fade
- Content scale from 0.95

Tabs:

- Animated underline slide
- Content fade on change

Progress bars:

- Animated fill with easing
- Subtle shimmer

Checkboxes:

- Scale + check draw animation

src/components/animations/confetti.tsx

- Triggered on achievements, perfect scores
- Canvas-based particles
- Short, celebratory

Prompt 42: Toast Notifications

Create toast notification system.

src/components/ui/toast.tsx

- Variants: success, error, warning, info
- Icon for each type
- Title and description
- Progress bar for auto-dismiss
- Close button
- Animated entrance/exit

src/components/ui/toaster.tsx

- Container at top-right
- Stacks multiple toasts
- Handles queue

src/lib/hooks/use-toast.ts

```
``typescript
```

```
const { toast } = useToast()
```

```
toast.success('Case submitted!', 'Your answer has been recorded')
toast.error('Error', 'Failed to load case')
toast.warning('Time running out', 'Only 2 minutes remaining')
toast.info('Tip', 'You can use hints in learning mode')
toast.dismiss(id)
...

```

```
## src/providers/toast-provider.tsx

```

- Wrap app
- Manage toast state
- Auto-dismiss after 5 seconds

Prompt 43: Empty States & Error Handling

Create comprehensive empty states and error handling.

```
## src/components/ui/empty-state.tsx

```

Props: icon, title, description, action

Variants:

- No cases found (search/filter)
- No progress yet (new user)
- No assessments available
- Search no results

```
## src/components/ui/error-state.tsx

```

Props: error, onRetry

- Friendly message
- "Try Again" button
- Report issue link

```
## src/app/not-found.tsx

```

- ARKA branding
- "Page not found" message
- "Back to Cases" link

- Search suggestion

src/app/error.tsx

- Global error boundary
- Friendly message
- Retry button
- Contact support link

src/components/ui/loading-state.tsx

- ARKA logo animation
- Progress text
- Skeleton options

Prompt 44: SEO & Metadata

Add comprehensive SEO and metadata.

src/app/layout.tsx

- Default metadata object:
 - Title template: "%s | ARKA-ED"
 - Description
 - Open Graph image
 - Twitter card
 - Favicon

Page-specific metadata:

- Landing: "Medical Imaging Education"
- Cases: "Practice Clinical Cases"
- Progress: "Track Your Learning"
- Pricing: "ARKA-ED Plans"

src/app/opengraph-image.tsx

- Dynamic OG image generation
- ARKA branding
- Key tagline

```
## src/app/sitemap.ts
```

- Marketing pages
- Exclude dashboard/admin

```
## src/app/robots.ts
```

- Allow marketing pages
- Disallow dashboard/admin

```
## JSON-LD Structured Data:
```

- Organization schema
- Educational application schema
- Course schema for specialty tracks

Prompt 45: Deployment & CI/CD

```
Prepare for production deployment.
```

```
## vercel.json
```

```
```json
{
 "framework": "nextjs",
 "regions": ["iad1"]
}
```
```

```
## next.config.js
```

- Image optimization
- Security headers
- Redirects if needed

```
## .env.production.example
```

- All production env vars with comments

```
## package.json scripts:
```

- build, start, lint, type-check

```
## .github/workflows/ci.yml
- Lint check
- Type check
- Build test

## Pre-deployment checklist page:
src/app/admin/deployment-check/page.tsx
- Environment vars check
- Database connection
- Auth configuration
- All systems status

## Security:
- RLS enabled on all tables
- Auth middleware verified
- No exposed secrets
- CSP headers
```

Appendices

Appendix A: Component Reference

| Path | Component | Purpose |
|--------------------------|-------------------|--------------------|
| ui/button | Button | Primary CTA |
| ui/card | Card | Content containers |
| ui/badge | Badge | Status indicators |
| cases/case-card | CaseCard | Library items |
| cases/ordering-interface | OrderingInterface | Imaging selection |
| cases/feedback-panel | FeedbackPanel | Post-submission |
| cases/acr-rating-badge | ACRRatingBadge | 1-9 display |

| | | |
|--------------------------|------------------|-------------|
| charts/progress-chart | ProgressChart | Line chart |
| charts/competency-radar | CompetencyRadar | Radar chart |
| layout/dashboard-sidebar | DashboardSidebar | Navigation |

Appendix B: Database Quick Reference

```
sql

-- Core tables
profiles (id, email, role, specialty_track)
cases (id, slug, title, category, difficulty)
imaging_options (id, name, modality, cost, radiation)
case_imaging_ratings (case_id, imaging_option_id, acr_rating)
user_case_attempts (user_id, case_id, score)
assessments (id, title, case_ids, time_limit)
user_assessments (user_id, assessment_id, score, status)
```

Appendix C: Cursor Best Practices

- 1. **Be Specific:** Include exact file paths and component names
- 2. **Provide Context:** Reference existing code
- 3. **One Task at a Time:** Break complex features into smaller prompts
- 4. **Include Types:** Specify TypeScript interfaces upfront
- 5. **Reference Design:** Describe exact visual requirements

Quality Checks After Each Feature:

```
bash

npm run build # Catch build errors
npm run lint # Code quality
# Manual browser testing
```

Mobile responsiveness check

Appendix D: Demo Script

1. **Landing Page** (30 sec) - Value proposition
2. **Registration** (30 sec) - Quick flow
3. **Case Library** (1 min) - Filtering, variety
4. **Single Case - Learning Mode** (3 min) - Full workflow
5. **Progress Dashboard** (1 min) - Analytics
6. **Assessment Mode** (1 min) - Timed quiz
7. **Institutional Value** (1 min) - Admin features

Total: 8-10 minutes + Q&A

Appendix E: Post-Launch Checklist

Content:

- ☐ 50+ cases loaded
- ☐ Medical accuracy reviewed
- ☐ ACR ratings verified
- ☐ Explanations thorough

Functionality:

- ☐ Registration/login works
- ☐ All cases load correctly
- ☐ Ordering interface intuitive
- ☐ Progress tracking updates
- ☐ Mobile polished

Performance:

- ☐ Page load < 2 seconds
- ☐ No console errors
- ☐ Images optimized

☐ Lighthouse > 90

Final Notes

What Makes This Demo "Launch Ready"

1. **Professional Polish:** Every interaction feels intentional
2. **Medical Accuracy:** Content reviewed and evidence-based
3. **Institutional Ready:** Admin features demonstrable
4. **Scalable Architecture:** Ready for more cases/features
5. **Performance:** Fast, responsive, reliable

Success Metrics

A successful demo should result in:

- Clear understanding of platform value
- Faculty engagement with content
- Questions about integration and pricing
- Request for pilot program

Next Steps After Demo

1. Gather pilot user feedback
 2. Expand case library
 3. Add LMS integration (LTI)
 4. Build institutional licensing
 5. Develop CME accreditation
-

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